Name:_____ Period:_____

Steal a Planet's Speed

Learning Intention: Today we will be learning how a planet's gravity can change a spacecraft's speed so we can explain how the gravity of other objects can change an orbit.

Success Criteria:
- I can use data to show how a flyby changes a spacecraft's speed.
- I can explain why passing behind a planet speeds a spacecraft up and passing in front slows it down.
- I can connect a gravity assist to a real mission.

Part 1: Predict

1. A spacecraft flies past a moving planet without using any fuel. What can the planet's gravity do to the spacecraft's speed? Circle one.

Speed it up Slow it down Both, depending on the path Nothing

Why? _____

Part 2: Investigate

Open the **Slingshot Challenge** from Canvas and stay on **1 · Investigate**. For each launch position below: set the position, choose your prediction in the tool, press Launch, and record what happened.

2. Record your data.

Launch position	Was my prediction right? (✓, ✗, or crashed)	Passed the planet (behind or in front)	Change in speed (km/s)
−6			
−4			
−2			
0			
+2			
+4			
+6			

3. Look at your data.

a. Which launch positions made the spacecraft speed up?

_____

b. Which launch positions made it slow down?

_____

4. Compare launch positions −2 and −6. Which one changed the speed more? What does that tell you about how close the spacecraft passed to the planet?

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5. What happened at launch position 0? Why?_____

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Part 3: How a Gravity Assist Works

Planets move fast as they orbit the Sun. When a spacecraft flies close to a planet, the planet's gravity pulls on it and bends its path. This is called a **gravity assist**.

- **Pass behind the planet** (the side it is moving away from), and the planet pulls the spacecraft along with it. The spacecraft **speeds up**.
- **Pass in front of the planet**, and the planet pulls back against the spacecraft's motion. The spacecraft **slows down**.

The planet gives up a tiny bit of its own motion, so small it can never be measured. No fuel is used.

6. Complete the pattern using your data.

Passing _____ the planet speeds a spacecraft up. Passing _____ the planet slows it down. The _____ the spacecraft passes, the bigger the change in speed.

Part 4: The Challenge

Real spacecraft use gravity assists all the time. In this challenge, you will fly three real missions.

- **Voyager 1 (NASA, launched 1977)** explored Jupiter and Saturn. In 1979 it flew past Jupiter to get a huge speed boost, and it is now the farthest human-made object from Earth.
- **Parker Solar Probe (NASA, launched 2018)** studies the Sun up close. It flew past Venus seven times to slow down, so it could fall closer and closer to the Sun.
- **Rosetta (European Space Agency, launched 2004)** was sent to catch comet 67P. No rocket was strong enough to send it straight there, so it flew past Earth three times and Mars once to build up speed, then reached the comet in 2014.

Your job: use what you learned in Part 2 to aim each spacecraft so it leaves the planet at exactly the target speed.

Click **2 · Challenge** and choose **Set A**. You get **one launch per mission**. Use your data from Part 2 to aim. A perfect flight (within 0.1 km/s of the target) scores 100 points.

7. Before you launch Mission 2 (Parker Solar Probe at Venus), which side of Venus should you aim for? Use your data to explain.

_____

8. Record your results.

Mission	Target speed (km/s)	Launch position I chose	Points
Voyager 1 at Jupiter			
Parker Solar Probe at Venus			
Rosetta at Earth			
		Total	/ 300

9. *Exit question:* In two sentences, explain how a spacecraft can change its speed without using fuel. Use evidence from your data.

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